Task-1

Leg 1 : Start → WP1

$\varphi = 38.418012°$ $\quad 38.418603°$

$\lambda = -110.777203°$ $\quad -110.775724$

$\Delta\varphi = 0.000591° = 1.0315 \times 10^{-5}$ rad

$\Delta\lambda = 0.001419° = 2.4766 \times 10^{-5}$ rad.

$\cos\varphi_1 . \cos\varphi_2 = \cos(38.418012°) \times \cos(38.418603°)$

$= 0.6139$

Haversine formula:

$\Delta\varphi/2 = 5.157 \times 10^{-6}$ $\quad \therefore \sin(\Delta\varphi/2) = 2.66 \times 10^{-11}$

$\Delta\lambda/2 = 1.2383 \times 10^{-5}$ $\quad \therefore \sin(\Delta\lambda/2) = 1.5334 \times 10^{-10}$

~~$a = 1.0988 \times 10^{-5}$~~

$a = 2.66 \times 10^{-11} + 9.413 \times 10^{-11}$

$= 1.2079 \times 10^{-10}$

$\sqrt{a} = 2.1976 \times 10^{-5}$

$\therefore d_1 = 6371000 \times 2.1976 \times 10^{-5} = 140$ m

Bearing:

North $= 6371000 \times 1.0315 \times 10^{-5} = 65.71\ m$

East $= 6371000 \times 2.4761 \times 10^{-5} \times 0.7835$

$= 157.79 \times 0.7835 = 123.63\ m$

$\sqrt{(65.71)^2 + (123.63)^2} = 140.\ m$

$\theta_1 = \text{atan}(123.63/65.71) = 62.0^\circ$

Leg 2: WP1 → WP2

$\Delta\varphi = 33.42016\overline{7} - 33.418603 = 0.001564^\circ = 2.7298 \times 10^{-5}$ rad

$\Delta\lambda = -110.776319 - (-110.775784) =$

$= -0.000535^\circ$

$= -9.337 \times 10^{-6}$ rad

$\cos\varphi_1 \cdot \cos\varphi_2 = 0.7835 \times 0.7835 = 0.6139$

Haversin :-

$\sin^2(\Delta\phi/2) = (1.3649 \times 10^{-5})^2 = 1.863 \times 10^{-10}$

$\sin^2(\Delta\lambda/2) = (4.668 \times 10^{-6})^2 = 2.180 \times 10^{-11} \times 0.6139$

$= 1.338 \times 10^{-11}$

$a = 1.863 \times 10^{-10} + 1.338 \times 10^{-11} = 1.997 \times 10^{-10}$

$\sqrt{a} = 2.8262 \times 10^{-5}$

$d_2 = 6371000 \times 2.7298 \times 10^{-5} = 173.91$

Bearing :-

North = $6371000 \times 2.7298 \times 10^{-5} = 173.91$ m

East = $6371000 \times (-9.337 \times 10^{-6}) \times 0.7835$

$= -59.49 \times 0.7835 = -46.61$ m

$\sqrt{(173.91)^2 + (46.61)^2} = 180.05$ m

$\text{atan}(46.61/173.91) = \text{atan}(0.2680) = 15^\circ$

$\theta = 360 - 15^\circ = 345^\circ$

Leg 3: WP2 → WP3

$\Delta\varphi = 0.000122^\circ = 2.129 \times 10^{-6}$ rad

$\Delta\lambda = 0.000073^\circ = 1.274 \times 10^{-6}$ rad

~~$\Delta\varphi = 2.129 \times 10^{-6}$ rad~~

product $= \cos\varphi_1 . \cos\varphi_2$

$= 0.6139$

Haversine:

$\sin^2(\Delta\varphi/2) = (1.0647 \times 10^{-6})^2 = 1.134 \times 10^{-12}$

$\sin^2(\Delta\lambda/2) = (6.37 \times 10^{-7})^2 = 4.06 \times 10^{-13} \times 0.6139$

$= 2.49 \times 10^{-13}$

$a = 1.134 \times 10^{-12} + 2.49 \times 10^{-13} = 1.383 \times 10^{-12}$

$\sqrt{a} = 2.352 \times 10^{-6}$

$d_3 = 6371000 \times 2.352 \times 10^{-6} = 14.98$

≈ 15 m

Bearing L

North = $6371000 \times 2.129 \times 10^{-6} = 13.57m$

East = $6371000 \times 1.274 \times 10^{-6} \times 0.7855$ ~~$\times 0.78$~~

$= 6.36m$

$\theta_3 = a\tan(6.36/13.57) = a\tan(0.4687)$

$= 25.1^\circ$

Turn Calculation:

command

Right 62°

1st turn = $62.0 - 0 = +62.0^\circ$

2nd " = $345.0 - 62.0 = +283^\circ$ left 77°

3rd " = $25.1 - 345 = -319.9^\circ$ Right 40.1°

command sequence :

(1) turn right 62° ⟶ turn 1

(2) drive 140 m ⟶ d_1

(3) turn left 77° ⟶ turn 2

(4) drive 180 m ⟶ d_2

⑤ turn right 40.1° → Turn 3

⑥ drive 15 m → d_3

Total path → 335 m

GPS ± 3m analysis

This, I don't know.

Iam sorry TwT